

Stavan Rupareliya

Design Verification / RTL / Embedded Systems

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PROFESSIONAL SUMMARY

Verification-focused engineer with 4.6+ years of experience in C/C++, Python, embedded Linux, hardware-software integration, debugging, and system validation. Builds auditable RTL and self-checking verification environments using reference models, assertions, boundary-focused stimulus, coverage goals, and reproducible CI. Experienced in translating system behavior into measurable verification evidence.

CORE SKILLS

Design & Verification: SystemVerilog, Verilog, RTL design, synchronous FIFO, scoreboards, assertions, constrained-random testing, functional coverage, boundary testing, FSM verification

Software & Systems: C/C++, Python, Embedded Linux, Linux, Git, GitHub Actions, Docker, GDB, Valgrind, hardware-software integration, performance profiling

AI / Embedded: OpenCV, GStreamer, PyTorch, TensorFlow, TFLite, edge inference, model optimization, multi-camera pipelines

FEATURED DESIGN VERIFICATION PROJECT

PARAMETERIZED SYNCHRONOUS FIFO - RTL & ASSERTION-DRIVEN VERIFICATION

SystemVerilog | Reference-model scoreboard | Assertions | Functional closure | GitHub Actions

- Designed synthesis-oriented, configurable RTL supporting arbitrary non-power-of-two depths, registered reads, programmable almost-full/almost-empty thresholds, and one-cycle overflow/underflow diagnostics.
- Defined and verified simultaneous-operation contracts at full and empty boundaries, including full-throughput pop/push behavior and non-fall-through empty semantics.
- Built an independent cycle-accurate scoreboard with directed boundary scenarios and 5,000 reproducible constrained-random cycles; checks ordering, data integrity, occupancy, flags, errors, and pointer rollover after every transaction.
- Added assertions and mandatory event-closure gates for full, empty, overflow, underflow, simultaneous transfers, and wraparound; automated regression and waveform retention with GitHub Actions.

github.com/stavanrupareliya36/verilog-programs/tree/master/fifo

PROFESSIONAL EXPERIENCE

TATA ELXSI LIMITED - Senior Software Engineer | Mar 2025 - Present

- Develop and optimize AI/ML text and image summarization pipelines for multimedia applications on embedded platforms.
- Profile, debug, and tune inference pipelines on resource-constrained embedded Linux devices, validating performance and stable system behavior.

EINFOCHIPS, AN ARROW COMPANY - Software Engineer | Jul 2021 - Mar 2025

- Designed real-time video analytics systems integrating AI models with embedded hardware; led hardware-software integration and end-to-end validation.
- Applied quantization and pruning to improve inference efficiency, and developed a modular camera simulator that accelerated PoC stimulus generation and validation.

EINFOCHIPS, AN ARROW COMPANY - Project Trainee Engineer | Jan 2021 - Jul 2021

- Completed hands-on C++, Python, embedded systems, Linux internals, system programming, and memory-management training; contributed to client PoCs.

ADDITIONAL TECHNICAL PROJECTS

MULTI-CAMERA SURVEILLANCE FOR AUTONOMOUS VEHICLES

- Designed and deployed driver-monitoring, lane-detection, and pedestrian-detection pipelines for four camera streams using GStreamer and OpenCV on i.MX95 NPU, achieving under 200 ms latency per frame.

VIDEO ANALYTICS SYSTEM USING EDGE COMPUTING

- Built and validated real-time C++/Python analytics for detection, recognition, counting, and safety use cases on edge hardware, reducing cloud dependency and supporting low-latency deployment.

SMART PARKING MANAGEMENT SYSTEM

- Designed and simulated Verilog FSM logic for entry/exit sequencing, slot accounting, gate control, and state-transition boundary cases.

TECHNICAL PUBLICATIONS

- Real-time detection and identification of plant leaf diseases using convolutional neural networks on an embedded platform - The Visual Computer, Springer, 2020.
- Real-time Face Mask Detection System on Edge using Deep Learning and Hardware Accelerators - IEEE C214, 2021.
- Real-Time Tomato Detection, Classification, and Counting System Using Deep Learning and Embedded Systems - Springer, 2021.
- Face Mask Detection and Counting Using Deep Learning and Embedded Systems - Advances in VLSI, Communication, and Signal Processing, Springer, 2022.

INDUSTRY ENGAGEMENT & RECOGNITION

- Delivered invited sessions at Nirma University on Embedded AI, Edge Computing, and Computer Vision for VLSI and Embedded Systems students and faculty.
- Core Value Award - Customer First, Jul 2024; Pat On The Back Award, Mar 2022.

EDUCATION

Master of Technology, Embedded Systems Engineering | Nirma University | 2018 - 2020

Bachelor of Engineering, Instrumentation and Control Engineering | Gujarat Technological University | 2012 - 2016